EXAMINATION NO: _____

SAINT GALLEN EXAMINATION BOARD

2018/2019 END OF **TERM TWO** EXAMINATION

FORM 3

MATHEMATICS

Subject Number: M131/II

Time Allowed: 2 h 30 mins

PAPER II
(100 marks)

Instructions

1. **This paper contains 13 printed pages. Please check.**
2. Answer **all six** questions in Section A and any **4** questions in Section B.
3. The maximum number of marks for each answer is indicated against each question.
4. Write your answers in the spaces provided on the question paper.
5. Calculators may be used.
6. **All working must be clearly shown.**
7. Write your **Examination Number** in the spaces provided on top of every page of the question paper.
8. In the table provided in this page, **tick** against the question number you have answered.

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
5			
6			
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9			
10			
11			
12			

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Section A (60 marks)

Answer **all** the **six** questions in this section

1. a. Simplify $\frac{x-1}{x} - \frac{2x+1}{3x}$ **(4 marks)**

b. A cylindrical container with diameter 7.9 cm has a volume of 343 cm^3. Calculate the height of the container, leaving your answer to one significant Figure. **(5 marks)**

2. a. Express n in terms of x and y given that $n \propto xy^{\frac{1}{2}}$ and $n = 2$ when $x = 1$ and $y = 4$. **(5 marks)**

b. Simplify $\frac{3ab-12ab^3}{1-2a}$ **(4 marks)**

Continued/ . . .

3. a. The angle of depression of two ships from the top of a tower are 30° and 60°. The ships are in a straight line from the point of observation. How far apart are the two ships if the tower is 400 m high. **(6 marks)**

b. The sum of interior angles of a regular polygon is 1440°, calculate the number of sides of the polygon. **(5 marks)**

Continued/ . . .

4. a. Atusaye mixed 500 kg of maize flour whose price is K90 per kg with 300 kg whose price is K70 per kg. What is the average price of 1 kg of the mixture? **(6 marks)**

b. Without using tables or a calculator, find the value of $\frac{\sin 60°}{\cos 30° + \sin 30°}$ **(5 marks)**

Continued/ . . .

5. a. Given that $\log_8 2x + \log_8(x-3) = 1$, find the value of x. **(6 marks)**

b. In **Figure 2**, **AC** is the diameter of a circle center **O.**

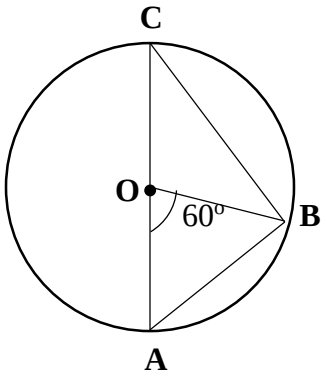


Figure 2

If angle **AOB** = 60°, calculate angle **CAB**. **(4 marks)**

6. a. Find the equation of a straight line that passes through (-1, 1) and (3, 2). **(5 marks)**

b. **Table 1** shows masses (m) in kg of different capacities (c) of a container in litres.

Table 1

$c\ (l)$	1	3	6	10	20
$m\ (kg)$	0.5	4.5	18	50	200

If m varies as the square of c, find the constant of proportionality, and hence the equation connecting m and c. **(5 marks)**

Section B (40 marks)
Answer any **four** questions from this section.

7. In a street of 150 houses, three different newspapers are delivered: The Nation, Daily Times and Malawi News. Of these, 40 receive The Nation, 35 receive Daily Times, and 60 receive Malawi News; 7 receive the Nation and Daily Times, 10 receive Daily Times and Malawi News and 4 the Nation and Malawi News; 34 receive no paper at all.

i. Draw a Venn diagram to represent this information.

ii. Use your Venn diagram to find how many houses receive all the three newspapers. **(10 marks)**

Continued/ . . .

8. A car starts from stage **A** and moves due north at 150 km/h. At the same time another car leaves from stage **B**, 150 km due east of stage **A**, and moves on a bearing of 340°. How fast will the second car have to travel if it is to meet the first car? **(10 marks)**

9. Using a scale of 2 cm to represent 2 units on both axes:

i. Draw triangle **A**(3,3), **B**(5,2), **C(**5.4) and triangle **A'**(3,7), **B'**(7,5), **C'**(7,9) on the same axes on the **graph paper** provided on **page 10**.

ii. Describe fully the transformation that maps triangle **ABC** onto triangle **A'B'C'** **(10 marks)**

Continued/ . . .

10. The cost (C) of producing cars is partly constant and partly varies as the number (N) of cars produced. If the cost of making 40 cars is K748 000 and that of making 100 cars is K1 420 000. Calculate the cost of making 200 cars. **(10 marks)**

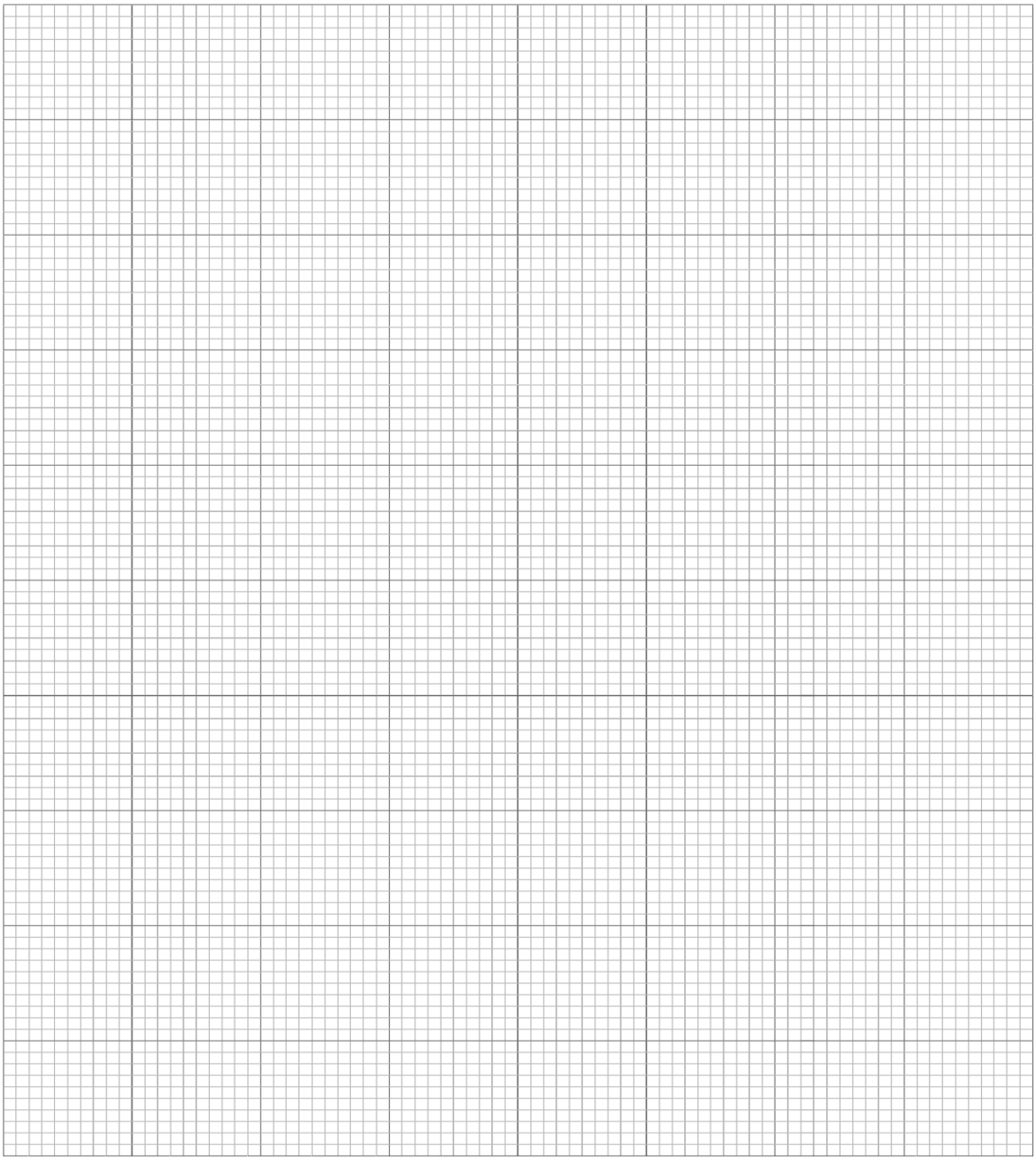
11. Solve the simultaneous equations:

$4m - n = 7$

$mn = 15$ **(10 marks)**

Continued/ . . .

12. Using a scale of 2 cm to represent 1 unit on both axes, draw the region represented by the following inequalities: $x + y < 1$, $y - x < 4$ and $2x + 3y \geq 6$. **(10 marks)**



END OF QUESTION PAPER

NB: This paper contains 13 printed pages.